

Formula sheet for Quiz 5

Note to students: you will be handed a copy with the quiz, so nothing here needs memorizing. It lists more than any one question needs, so deciding which applies is still your job.

A layer, forward, and shapes

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}, \quad \mathbf{a} = \phi(\mathbf{z}), \quad \mathbf{W} : (n_{\text{out}} \times n_{\text{in}}), \quad \mathbf{b} : (n_{\text{out}})$$

The **(units, inputs)** convention, unchanged since Week 2. A gradient has the same shape as the thing it differentiates.

Broadcasting

Shapes align **from the right**; a missing axis counts as size 1, and any axis of size 1 stretches to match. Two axes that are neither equal nor 1 are an error.

$$(N, 1) \text{ with } (N,) \rightarrow (N, 1) \text{ with } (1, N) \rightarrow \text{result } (N, N)$$

Batching

$$\text{iterations per epoch} = \left\lceil \frac{N}{B} \right\rceil \quad \text{standard error of the gradient} \propto \frac{1}{\sqrt{B}}$$

Losses

$$\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2 \quad \text{BCE} = -\frac{1}{n} \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

For C classes with scores $z_1, \dots, z_C$ and true class y : $\text{softmax } p_j = e^{z_j} / \sum_k e^{z_k}$, negative log-likelihood $\ell = -\log p_y$.

Optimizer updates

With $g_t = \nabla_{\theta} \mathcal{L}(\theta_t)$, rate η , and $\epsilon = 10^{-8}$:

SGD $\theta_{t+1} = \theta_t - \eta g_t$	Momentum $v_{t+1} = \beta v_t + g_t, \quad \theta_{t+1} = \theta_t - \eta v_{t+1}$
RMSProp $s_{t+1} = \rho s_t + (1 - \rho) g_t^2, \quad \theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{s_{t+1} + \epsilon}} g_t$	Adam $m_{t+1} = \beta_1 m_t + (1 - \beta_1) g_t$ $v_{t+1} = \beta_2 v_t + (1 - \beta_2) g_t^2$ $\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{v_{t+1} + \epsilon}} m_{t+1}$

Defaults $\beta = 0.9$; $\beta_1 = 0.9$, $\beta_2 = 0.999$. Adam also divides m and v by $1 - \beta^t$ to correct their bias early in training.

Schedules, clipping, initialization, weight decay

$$\text{step } \eta_t = \eta_0 \gamma^{\lfloor t/s \rfloor} \quad \text{exponential } \eta_t = \eta_0 e^{-kt} \quad \text{cosine } \eta_t = \frac{\eta_0}{2} (1 + \cos \frac{\pi t}{T})$$

Clip by **norm** with threshold C , over all parameters together; clip by **value** clamps each component to $[-C, C]$.

$$g \leftarrow g \cdot \frac{C}{\|g\|} \text{ if } \|g\| > C \quad \text{Xavier } \text{Var}(W) = \frac{2}{n_{\text{in}} + n_{\text{out}}} \quad \text{He } \text{Var}(W) = \frac{2}{n_{\text{in}}}$$
$$\mathcal{L}_{\text{total}} = \mathcal{L} + \lambda \sum_j w_j^2, \quad \nabla_w \left[\lambda \sum_j w_j^2 \right] = 2\lambda w$$

Notation

η is the learning rate, λ the regularization strength, θ all parameters, ℓ one example's loss and $\mathcal{L}$ the cost. N is the training set size, B the batch size, C the number of classes, except in the clipping formula where C is the threshold.